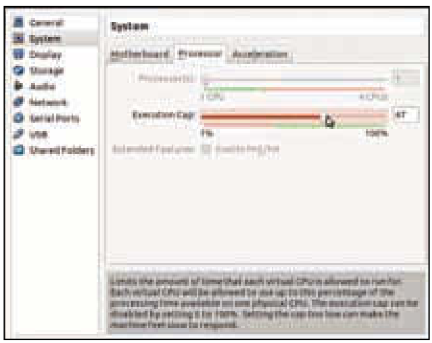
server, a third hosting the FTP server and yet another handling the emailing job and so on. At such a point in time, the following come into the picture:

- Purchasing as many machines as you require. This is going to get costly.
- Setting them up, connecting them to the network and installing software on them. If you're going to make two or more servers do the same job (such as two servers being used as a database server at one time), you're going to have to repeat the same task on both of them.
- Administering and maintaining them individually.
- Providing the extra electricity, networking facility and floor space (the space the server will occupy in the real world – think of it as the space you need to place your desktop's CPU cabinet in your room).

All of this might seem inevitable but with virtualization, it becomes a lot less complicated than it would be without it. With virtualization, the scene becomes something like this:

- You can create multiple virtual machines on the same server, thus saving on the cost of buying new machines. Also, you're making optimum use of one machine which is a good thing!
- Since you can set up a single virtual machine and then clone it to get the same setup again, setting up multiple servers repetitively becomes a thing of past – you're now in need of doing copy-paste rather than reconfiguration. This saves time, and adding new machines is much faster and easier.



Virtualbox dynamic guest processor frequency adjustment

- Advanced virtualization software for servers come with inbuilt monitoring and maintenance features, which not only save plenty of money and time but also lots of manual effort (and that's the best part).
- Since you're using only one server for running many machines (virtual machines), you don't need more electricity and floor space either. Notice that we just compared the scene with and without virtualization. There are other benefits which virtualization provides; a few follow: